

For the group assignment, build the below 5 models to predict the label of an image from a given image on the yelp Photos dataset

1. Use a pretrained VGG model for transfer learning by adding custom top layers. Incorporate dense layers that take in additional non-image features (e.g., business-related metadata), concatenate them with features extracted from the VGG base, and fine-tune the model end-to-end.
2. Build a model using transfer learning with a pretrained VGG network as the base. Add custom layers on top, including dense layers that incorporate additional features (e.g., business or photo metadata), and fine-tune the entire model.
3. Use a pretrained EfficientNet model for transfer learning. Unfreeze select layers of the base network and fine-tune the model to adapt it to the target task.
4. Use a pretrained VGG model for transfer learning. Unfreeze certain layers of the base network and apply fine-tuning to adapt the model to the new dataset.
5. Initialize a model using a pretrained VGG network and train all layers from the start without freezing any part of the base model.

Model Building and Evaluation - 20%

Correct Model Built with Confusion Matrix and AUC - 10%

2 Rounds of iterative model development to build data centric AI with interpretability - 10%

In transfer learning, we typically follow a two-step process. First, is the **feature extraction/transfer learning step**. The second step is **fine-tuning**. Fine-tuning should not be confused with **model or hyperparameter tuning**.

Submit a table at the beginning of your code that calls out the following for each model (double-check to make sure the table is accurate)

Task	Sub-task	Comments
Data Preprocessing	Scaling and Resizing	Done/Not Done
	Image Augmentation	Done/Not Done
	Train and test data handled correctly	Done/Not Done
	Gaussian Blur, Histogram Equalization and Intensity thresholds	Done/Not Done

Task	Sub-task	Comments
Model Trained	Training Time?	Done/Not Done
	AUC and Confusion Matrix Computed	Done/Not Done
	Overfitting/Underfitting checked and handled	Done/Not Done
Data Centric AI Development	Interpretability Implemented	Major Findings
	1st Round of Tuning	Issue Faced/Fixed
	2nd Round of Tuning	Issue Faced/Fixed